Topic : Random variables

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Random variable

Definition

Let $\mathcal{S}$ be a sample space associated with a random experiment E, then a random variable X is a real valued function defined on $\mathcal{S}$.
i.e., $X : \mathcal{S} \rightarrow \mathcal{R}$ such that for each element $s \in \mathcal{S}$, there is a unique real number $X = X(s)$ associated.

Example 1

Consider the random experiment of tossing a coin two times in succession. Then the sample space is $\mathcal{S} = \{HH, HT, TH, TT\}$.
Let X denotes the number of heads obtained, then X is a random variable and for each outcome, its value is given as,

$$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0.$$

In this case we can write, the range of X is $\{0, 1, 2\}$.

Example 2

A person plays a game of tossing a coin thrice. For each head, he is given Rs 2 by the organizer of the game and for each tail, he has to give Rs 1.50 to the organizer. Let X denote the amount gained or lost by the person. Then X is a random variable.

Here, $\mathcal{S} = \{HHH, HTT, THH, TTH, THT, TTT, HHT, HTH\}$. Then,
$X(HHH) = 3 \times 2 = 6$
$X(HTT) = X(THT) = X(TTH) = (1 \times 2) - (2 \times 1.50) = -1$
$X(HHT) = X(HTH) = X(THH) = (2 \times 2) - (1 \times 1.50) = 2.50$
$X(TTT) = 3 \times 1.50 = 4.50.$
Hence the range set of the random variable X is $\{-1, -2.50, 4.50, 6\}$.

There are two types of one dimensional random variables;

1. Discrete random variable
2. Continuous random variable

Discrete random variable

A random variable X is said to be **discrete** if X assumes only finite number of values or countably infinite values.

Example

The number of heads in four tosses of a coin is a discrete random variable. Because, here X takes only the values $0, 1, 2, 3, 4$.

Continuous random variable

A random variable X is said to be **continuous** if X assumes any value within an interval.

Example

The weights of a group of individuals in a class.

Discrete probability distribution

Let X be a discrete random variable which assumes values $x_1, x_2, x_3, ..., x_n, ...$ With each possible x_i we associate a real number $p_i = P(X = x_i)$ for $i = 1, 2, 3, ..., n, ...,$ satisfies the conditions

1. $p_i \geq 0$ for all $i = 1, 2, 3, ..., n, ...$
2. $\sum_{i=1}^{\infty} p_i = 1.$

Then the function p_i is called the **discrete probability distribution** or **probability mass function (p.m.f.)** or **probability density function** of the random variable X.

We can represent the probability mass function of X in a tabular form as below,

X	x_1	x_2	x_3	...	...	x_i	...	...	x_n	...
$p_i = P(X = x_i)$	p_1	p_2	p_3	...	...	p_i	...	...	p_i	...

Note: For a specified $t = x_i$,

$$P(X \leq t) = P(X = x_1) + P(X = x_2) + ... + P(X = x_{i-1}) + P(X = x_i)$$

$$P(X < t) = P(X = x_1) + P(X = x_2) + P(X = x_3) + ... + P(X = x_{i-1})$$

Distribution function or Cumulative distribution function (c.d.f.)

Let X be a any random variable. Then the **distribution function or cumulative distribution function (c.d.f.)** of X is a function F defined by

$$F(t) = P(X \leq t).$$

For a discrete random variable X, the c.d.f. of X is,

$$F(t) = P(X \leq t) = \sum_{x \leq t} P(X = x).$$

Expectation of a discrete random variable

Let X be a discrete random variable which assumes values $x_1, x_2, x_3, ..., x_n, ...$ with p.m.f. $p_i = P(X = x_i)$. Then the **expectation or mean** of X is defined as

$$E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i).$$

Similarly,

$$E(X^2) = \sum_{i=1}^{\infty} x_i^2 P(X = x_i).$$

Continuous probability distribution

Let X be a continuous random variable which assumes values from an interval $I \subseteq \mathbb{R}$. If there exists a function f satisfies the conditions;

1. $f(x) \geq 0$ for all $x \in I$
2. $\displaystyle\int_{-\infty}^{\infty} f(x)\, dx = 1.$

Then the function f is called the **continuous probability distribution** or **probability distribution function (p.d.f.)** of the random variable X.

Note: Let X be a continuous random variable. Then for a specific a, b such that $-\infty < a < b < \infty$ we have

$$P\left(a \leq x \leq b\right) = P\left(a < x \leq b\right) = P\left(a \leq x < b\right) = P\left(a < x < b\right) = \int_a^b f(x)\, dx.$$

For a continuous random variable X, the c.d.f. of X is,

$$F(t) = P(X \leq t) = \int_{-\infty}^{t} f(x)\, dx.$$

Expectation of a continuous random variable

Let X be a continuous random variable with p.d.f. $f(x)$. Then the **expectation or mean** of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx.$$

Similarly,

$$E(X^2) = \int_{-\infty}^{\infty} x^2\, f(x)\, dx.$$

Note: If X and Y are two random variables and α, β are two constants then $E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y)$ and
$E(\alpha X + \beta) = \alpha E(X) + \beta$.

Variance

Let X be a random variable with the probability distribution. Then the **variance** of X is defined as

$$V(X) = E\left[X - E(X)\right]^2$$

i.e.,

$$\begin{aligned}
V(X) &= E\left[X - E(X)\right]^2 \\
&= E\left[X^2 - 2XE(X) + (E(X))^2\right] \\
&= E(X^2) - 2E(X)E(X) + -\left[E(X)\right]^2 \\
&= E(X^2) - 2\left[E(X)\right]^2 + \left[E(X)\right]^2 \\
&= E(X^2) - \left[E(X)\right]^2
\end{aligned}$$

The **standard deviation** of X is defined as $S.D.(X) = \sqrt{V(X)}$.

Note: If X and Y are two random variables and α, β are two constants then

$$V(\alpha X + \beta Y) = \alpha^2 V(X) + \beta^2 V(Y)$$

and

$$V(\alpha X + \beta) = \alpha^2 V(X).$$

Problems

Question 1.

Five defective bulbs are accidently mixed with 20 good ones. It is not possible to just look at a bulb and tell whether or not it is defective. Find the probability distribution of the number of defective bulbs, if 4 bulbs are drawn at random from this lot?

Solution: Let X be the random variable denotes the number of defective bulbs out of 4. Then, X takes the values $0, 1, 2, 3, 4$.
Number of defective bulbs $=$5.
Number of good bulbs$=$20.
Total number of bulbs $=$ 25.

$$P(X=0) = P(\text{no defective bulb}) = P(\text{all good ones}) = \frac{20C_4}{225C_4} = \frac{969}{2530}.$$

$$P(X=1) = P(1 \text{ defective bulb and 3 good ones}) = \frac{5C_1 \times 20C_3}{225C_4} = \frac{1140}{2530}.$$
$$P(X=2) = P(2 \text{ defective bulb and 2 good ones}) = \frac{5C_2 \times 20C_2}{225C_4} = \frac{380}{2530}.$$
$$P(X=3) = P(3 \text{ defective bulb and 1 good ones}) = \frac{5C_3 \times 20C_1}{225C_4} = \frac{40}{2530}.$$
$$P(X=4) = P(4 \text{ defective bulb }) = \frac{5C_4}{225C_4} = \frac{1}{2530}.$$

Therefore the required p.m.f. is,

X	0	1	2	3	4
$P(X=x)$	$\frac{969}{2530}$	$\frac{1140}{2530}$	$\frac{380}{2530}$	$\frac{40}{2530}$	$\frac{1}{2530}$

Question 2.

The probability mass function (p.m.f.) of a random variable X is given by the following table:

X	-2	-1	0	1	2	3
$P(X=x)$	0.3	k	0.2	$2k$	0.3	k

1. Determine the value of k.
2. Find the mean and variance of X.
3. Find $P(X<1)$, $\quad P(-1 \leq X < 2)$.

Solution: (1) For a valid p.m.f. we have, $\sum_x P(X=x) = 1$ implies $0.6 + 4k = 1 \implies k = 0.1$.

(2)

$$\begin{aligned}\text{Mean of } X = E(X) &= \sum_{i=1}^{6} x_i P(X = x_i)\\ &= -2(0.1) - 1(k) + 0 + 1(k) + (0.3) + 3(k)\\ &= 0.4 + 4k = 0.8.\end{aligned}$$

$$
\begin{aligned}
E(X^2) &= \sum_{i=1}^{6} x_i^2 P(X = x_i) \\
&= 4(0.1) + 1(k) + 0 + 1(k) + (0.3) + 9(k) \\
&= 1.6 + 12k = 1.8.
\end{aligned}
$$

Therefore,variance of X is,

$$
\begin{aligned}
V(X) &= E(X^2) - (E(X))^2 \\
&= 1.8 - 0.64 = 2.16
\end{aligned}
$$

(3) $P(X < 1) = P(X = -2) + P(X = -1) + P(X = 0) = 0.5 + k = 0.6$
$P(-1 \leq X < 2) = P(X = -1) + P(X = 0) + P(X = 1) = 0.2 + 3k = 0.5$

Question 3.

A box contains 12 balls of which 3 are white and 9 are red. A sample of 3 balls is selected from the box. Let X denote the number of white balls in the sample. Find the p.m.f. of X. Determine the mean and standard deviation of X.

Solution: Here $X = 0, 1, 2, 3$. Then,

$$P(X=0) = \frac{9C_3}{12C_3} = \frac{84}{220}; \quad P(X=1) = \frac{3C_1 \times 9C_2}{12C_3} = \frac{108}{220}$$

$$P(X=2) = \frac{3C_2 \times 9C_1}{12C_3} = \frac{27}{220}; \quad P(X=3) = \frac{3C_3}{12C_3} = \frac{1}{220}$$

Therefore the required p.m.f. is,

X	0	1	2	3
$P(X=x)$	$\frac{84}{220}$	$\frac{108}{220}$	$\frac{27}{220}$	$\frac{1}{220}$

Question 4.

Let X be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} 3x^2, & \text{if } 0 < x < 1, \\ 0, & \text{else where} \end{cases}$$

Then find $E(X)$ and $V(X)$?

Solution: We have,

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x\ f(x)\ dx \\ &= \int_0^1 x\ (3x^2)\ dx \\ &= 3\int_0^1 x^3\ dx \\ &= \frac{3}{4}. \end{aligned}$$

Also,

$$
\begin{aligned}
E(X^2) &= \int_{-\infty}^{\infty} x^2 \ f(x) \ dx \\
&= \int_0^1 x^2 \ (3x^2) \ dx \\
&= 3 \int_0^1 x^4 \ dx \\
&= \frac{3}{5}.
\end{aligned}
$$

Therefore, $V(X) = E(X^2) - (E(X))^2 = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}$.

Question 5.

Let X be a continuous random variable with p.d.f.

$$f(x) = \; ke^{-|x|}, \quad \text{if } -\infty < x < \infty \;.$$

Then find the value of k, $E(X)$ and $S.D.(X)$?

Solution: We have, $|x| = \begin{cases} -x, & \text{if } x < 0, \\ x, & \text{if} x \geq 0 \end{cases}$ Therefore, the p.d.f. becomes, $f(x) = \begin{cases} ke^{x}, & \text{if } x < 0, \\ ke^{-x}, & \text{if } x \geq 0 \end{cases}$

Since $f(x)$ is a valid p.d.f, we've

$$\int_{-\infty}^{\infty} f(x)\, dx = 1 \implies \int_{-\infty}^{\infty} \underbrace{ke^{-|x|}}_{\text{even function}} \quad dx = 1$$

$$\implies 2\int_{0}^{\infty} ke^{|x|}\, dx = 1 \implies 2k\int_{0}^{\infty} e^{-x}\, dx = 1$$

$$\implies 2k\left(-e^{-x}\right)_{0}^{\infty} = 1 \implies k = \tfrac{1}{2}.$$

We have,

$$
\begin{aligned}
E(X) &= \int_{-\infty}^{\infty} x \; f(x) \; dx \\
&= \int_{-\infty}^{\infty} \underbrace{xe^{-|x|}}_{\text{odd function}} \; dx = 0
\end{aligned}
$$

Also,

$$
\begin{aligned}
E(X^2) &= \int_{-\infty}^{\infty} x^2 \; f(x) \; dx \\
&= \int_{-\infty}^{\infty} \underbrace{x^2 e^{-|x|}}_{\text{even function}} \; dx \\
&= 2\int_{0}^{\infty} x^2 e^{-|x|} \; dx = 2\int_{0}^{\infty} x^2 e^{-x} \; dx = 2
\end{aligned}
$$

So, $V(X) = E(X^2) - (E(X))^2 = 2$. Hence, $S.D.(X) = \sqrt{2}$.

Question 6.

Let X be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} x, & \text{if } 0 \le x \le 1, \\ 2-x, & 1 < x \le 2 \\ 0, & \text{else where} \end{cases}$$

Then find the c.d.f of X and find (i) $P\left(X \ge \frac{3}{2}\right)$ (ii) $P\left(\frac{3}{2} < X \le 2\right)$?

Solution: We know the c.d.f of X is $F(t) = P(X \le t) = \int_{-\infty}^{t} f(x)\, dx$.

Case(1): When $-\infty < t < 0$

We have, $F(t) = \int_{-\infty}^{t} f(x)\, dx = \int_{-\infty}^{t} 0\, dx = 0.$

Case(2): When $0 \le t \le 1$

We have, $F(t) = \int_{-\infty}^{t} f(x)\, dx = \int_{-\infty}^{0} f(x)\, dx + \int_{0}^{t} f(x)\, dx =$

$\int_{-\infty}^{0} 0\, dx + \int_{0}^{t} x\, dx = \frac{t^2}{2}.$

Case(3): When $1 < t \leq 2$

We have,

$$F(t) = \int_{-\infty}^{t} f(x)\ dx = \int_{-\infty}^{0} f(x)\ dx + \int_{0}^{1} f(x)\ dx + \int_{1}^{t} f(x)\ dx =$$

$$\int_{-\infty}^{0} 0\ dx + \int_{0}^{1} x\ dx + \int_{1}^{t} (2-x)\ dx = \frac{1}{2} + \left(-\frac{(t-2)^2}{2} + \frac{1}{2}\right) = 1 - \frac{(t-2)^2}{2}.$$

Case(4): When $t \geq 2$

We have, $F(t) = \int_{-\infty}^{t} f(x)\ dx =$

$$\int_{-\infty}^{0} f(x)\ dx + \int_{0}^{1} f(x)\ dx + \int_{1}^{2} f(x)\ dx + \int_{1}^{t} f(x)\ dx =$$

$$\int_{-\infty}^{0} 0\ dx + \int_{0}^{1} x\ dx + \int_{1}^{t} (2-x)\ dx \int_{1}^{t} 0\ dx = 0 + \frac{1}{2} + \frac{1}{2} + 0 = 1.$$

Hence the required c.d.f of X is,

$$F(t) = P(X \leq t) = \begin{cases} 0, & \text{if } -\infty < t < 0, \\ \frac{t^2}{2}, & \text{if } 0 \leq t \leq 1 \\ 1 - \frac{(t-2)^2}{2}, & \text{if } 1 < t \leq 2 \\ 0, & \text{if } t > 2 \end{cases}$$

(i) We know that,

$$\begin{aligned} P(X \geq \frac{3}{2}) &= 1 - P(X < \frac{3}{2}) \\ &= 1 - F\left(\frac{3}{2}\right) \\ &= 1 - \left(1 - \frac{\left(\frac{3}{2} - 2\right)^2}{2}\right) = \frac{1}{8}. \end{aligned}$$

Or,

$$
\begin{aligned}
P(X \geq \frac{3}{2}) &= 1 - P(X < \frac{3}{2}) \\
&= 1 - \int_{-\infty}^{\frac{3}{2}} f(x)\ dx \\
&= 1 - \int_{-\infty}^{0} 0\ dx + \int_{0}^{1} x\ dx + \int_{1}^{\frac{3}{2}} (2-x)\ dx = \frac{1}{8}.
\end{aligned}
$$

(ii) We have $P\left(\frac{3}{2} < X \leq 2\right) = F(2) - F\left(\frac{3}{2}\right) = 1 - \frac{7}{8} = \frac{1}{8}$.

Or,

$$
\begin{aligned}
P\left(\frac{3}{2} < X \leq 2\right) &= \int_{\frac{3}{2}}^{2} f(x)\ dx \\
&= \int_{\frac{3}{2}}^{2} (2-x)\ dx \\
&= \left(2x - \frac{x^2}{2}\right)_{\frac{3}{2}}^{2} \frac{1}{8}.
\end{aligned}
$$

Two dimensional random variable

Let S be the sample space associated to a random experiment. Let X and Y be two random variables on S. Let $X = f(s)$ and $Y = g(s)$ for $s \in S$ where f and g are real valued functions defined on S. Then the pair (X, Y) is called the **two dimensional random variable**.

Two dimensional discrete random variable: Let (X, Y) be a two dimensional random variable. If (X, Y) takes finite or countably infinite values then (X, Y) is called a two dimensional discrete random variable. In this case we can write,

$$(X, Y) = \Big\{(x_i, y_j) \;\mid\; i = 1, 2, 3, ..., n, ... \;\&\; j = 1, 2, 3, ..., m, ...\Big\}$$

Two dimensional continuous random variable: Let (X, Y) be a two dimensional random variable. If (X, Y) assumes values in an uncountable set of the Euclidean plane $\mathbb{R}^2$ then (X, Y) is called a two dimensional continuous random variable. In this case we can write,

$$(X, Y) = \Big\{(x, y) \;\mid\; (x, y) \in R \subseteq \mathbb{R}^2\Big\}$$

Discrete joint probability distribution

Let (X, Y) be the two dimensional discrete random variable. With each possible outcome (x_i, y_j) for $i = 1, 2, 3, \ldots n, \ldots$ and $j = 1, 2, 3, \ldots m, \ldots$ we associate a real number $p_{ij} = p(x_i, y_j) = P\left(X = x_i, Y = y_j\right)$ satisfying the conditions:

1. $p_{ij} \geq 0$ for all $i = 1, 2, 3, \ldots n, \ldots$ and $j = 1, 2, 3, \ldots m, \ldots$
2. $\displaystyle\sum_{j=1}^{\infty}\sum_{i=1}^{\infty} p_{ij} = 1$

Then p_{ij} is called the joint probability density function of (X, Y).

The discrete joint probability distribution can be represented in the following table

$X \setminus Y$	y_1	y_2	y_3	...	y_j	...
x_1	p_{11}	p_{12}	p_{13}	...	p_{1j}	...
x_2	p_{21}	p_{22}	p_{23}	...	p_{2j}	...
x_3	p_{31}	p_{32}	p_{33}	...	p_{3j}	...
...	...	...	...	...	...	...
x_i	p_{i1}	p_{i2}	p_{i3}	...	p_{ij}	...
...	...	...	...	...	...	...

Marginal probability distributions

Let (X, Y) be the two dimensional discrete random variable. Then the marginal probability distribution of X is $P(X = x_i) = \sum_{j=1}^{\infty} p_{ij}$ for fixed i.

Similarly, the marginal probability distribution of Y is $P(Y = y_j) = \sum_{i=1}^{\infty} p_{ij}$ for fixed j.

Expectation

Let (X, Y) be the two dimensional discrete random variable with joint probability density function $p_{ij} = p(x_i, y_j)$ then the

1. Expectation or mean of X is $E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i)$
2. Expectation or mean of Y is $E(Y) = \sum_{j=1}^{\infty} y_j P(Y = y_j)$
3. Expectation or mean of XY is $E(XY) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x_i y_j p_{ij}$

Continuous joint probability distribution

Let (X, Y) be the two dimensional continuous random variable assuming values from a region $R \subseteq \mathbb{R}^2$. The joint probability distribution function $f(x, y)$ is a function satisfying the conditions:

1. $f(x, y) \geq 0$ for all $(x, y) \in R$
2. $\iint_R f(x, y)dxdy = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y)dydx = 1$

Note: Let (X, Y) be the two dimensional continuous random variable and if $-\infty < a < b < \infty$ and $-\infty < c < d < \infty$ then

1. $P(a \leq X \leq b, c \leq Y \leq d) = \int_{x=a}^{b} \int_{y=c}^{d} f(x, y)dydx$
2. $P(a \leq X \leq b) = \int_{x=a}^{b} \int_{y=-\infty}^{\infty} f(x, y)dydx$
3. $P(c \leq Y \leq d) = \int_{x=-\infty}^{\infty} \int_{y=c}^{d} f(x, y)dydx$

Marginal probability distributions

Let (X, Y) be the two dimensional continuous random variable with joint p.d.f. $f(x, y)$ then

1. the **marginal distribution of** X is
$$g(x) = P(X = x) = \int_{y=-\infty}^{\infty} f(x, y)dy \text{ for } -\infty < x < \infty.$$
2. the **marginal distribution** of Y is
$$h(y) = P(Y = y) = \int_{x=-\infty}^{\infty} f(x, y)dx \text{ for } -\infty < y < \infty.$$

Expectation

Let (X, Y) be the two dimensional continuous random variable with joint probability distribution $f(x, y)$ then the

1. Expectation of X is,
$$E(X) = \int_{x=-\infty}^{\infty} x\ g(x)\ dx = \int_{x=-\infty}^{\infty} x\ \left(\int_{y=-\infty}^{\infty} f(x, y) dy \right) dx$$
2. Expectation of Y is,
$$E(Y) = \int_{x=-\infty}^{\infty} y\ h(y)\ dy \int_{y=-\infty}^{\infty} y\ \left(\int_{x=-\infty}^{\infty} f(x, y) dx \right) dy$$
3. Expectation of XY is, $E(XY) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) dydx$

Problems

Question 1.

Let (X, Y) be the two dimensional random variable with joint probability distribution $p(x, y) = c(2x + y)$ where x and y can assume all integral values such that $0 \leq x \leq 2, 0 \leq y \leq 3$ and $p(x, y) = 0$ otherwise. Then find

1. the value of c.
2. $P(X = 2, Y = 1), P(X \geq 1, Y \leq 2), P(X + Y \leq 1)$

Solution: Here, $X = 0, 1, 2$ and $Y = 0, 1, 2, 3$. The joint probability distribution of (X, Y) is

$X \setminus Y$	0	1	2	3
0	0	c	$2c$	$3c$
1	$2c$	$3c$	$4c$	$5c$
2	$4c$	$5c$	$6c$	$7c$

(1) For a valid p.d.f., we have,

$$\sum_{i=1}^{3}\sum_{j=1}^{4} p\left(x_i, y_j\right) = 1 \implies 42c = 1 \implies c = \frac{1}{42}.$$

(2) We have, $P(X = 2, Y = 1) = 5c = \frac{5}{42}$.

$$\begin{aligned} P(X \geq 1, Y \leq 2) &= P(X = 1. Y \leq 2) + P(X = 2, Y \leq 2) \\ &= P(1,0) + P(2,0) + P(1,1) + P(2,1) + P(1,2) + P(2,2) \\ &= 24c = \frac{1}{2}. \end{aligned}$$

and

$$\begin{aligned} P(X + Y \leq 1) &= P(X = 0. Y = 1) + P(X = 1, Y = 0) \\ &= 3c = \frac{1}{14}. \end{aligned}$$

Question 2.

A coin is tossed three times. Let $X = 0$ or 1 according as a head or a tail occurs on the first toss. Let Y be equal to the total number heads occur. Then find

1. the joint probability distribution of (X, Y).
2. the marginal probability distribution of X and Y.
3. $E(X), E(Y), E(2X+Y), E(XY)$.

Solution: We have,
$\mathcal{S} = \left\{HHH, HHT, HTH, THH, THT, HTT, TTH, TTT\right\}$. Here $X = 0, 1$ and $Y = 0, 1, 2, 3$.

(1) The joint probability distribution of $(X.Y)$ is,

$X \setminus Y$	0	1	2	3
0	$P(X=0, Y=0) = 0$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
1	$P(X=1, Y=0) = \frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0

(2) The marginal probability distribution of X is,

X	0	1
$P(X=x)$	$\frac{4}{8}$	$\frac{4}{8}$

The marginal probability distribution of X is,

Y	0	1	2	3
$P(Y=y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(3) We have,

$$E(X) = \sum_{i=1}^{2} x_i P(X = x_i) = 0 \times P(X = 0) + 1 \times P(X = 1) = \frac{1}{2}.$$ and

$$\begin{aligned} E(Y) &= \sum_{j=1}^{4} y_j P(Y = y_j) \\ &= 0 \times P(Y = 0) + 1 \times P(Y = 1) + 2 \times P(Y = 2) + 3 \times P(Y = 3) \\ &= \frac{3}{2}. \end{aligned}$$

We have, $E(2X+Y) = 2E(X) + E(Y) = 1 + \frac{3}{2} = \frac{5}{2}$. and

$$
\begin{aligned}
E(XY) &= \sum_{i=1}^{2}\sum_{j=1}^{4} x_i y_j \ p_{ij} \\
&= (0x0) \times P(0,0) + (0x1) \times P(0,1) + (0x2) \times P(0,2) \\
&+ (0x3) \times P(0,3) + (1x0) \times P(1,0) + (1x1) \times P(1,1) \\
&+ (1x2) \times P(1,2) + (1x3) \times P(1,3) = \frac{1}{2}.
\end{aligned}
$$

Question 2.

Let (X, Y) be the two dimensional random variable with joint probability distribution $f(x,y) = \begin{cases} k(x+1)e^{-y}, & \text{if } 0 < x < 1, y > 0 \\ 0, & \text{else where} \end{cases}$. Then find

1. the value of k.
2. The marginal probability distributions of X and Y

Solution: We know that for a valid p.d.f., $\int_{x=-\infty}^{\infty}\int_{y=-\infty}^{\infty} f(x,y)dydx = 1$

$$\implies \int_{x=0}^{1}\int_{y=0}^{\infty} k(x+1)e^{-y}\ dydx = 1$$

$$\implies k\int_{x=0}^{1}(x+1)\left(\frac{e^{-y}}{-1}\right)_{y=0}^{\infty}\ dx = 1 \implies k\int_{x=0}^{1}(x+1)\ dx = 1$$

$$\implies k\left(\tfrac{x^2}{2}+x\right)_0^1 = 1 \implies \tfrac{3k}{2} = 1 \implies k = \tfrac{2}{3}.$$

The marginal probability distribution of X is, $g(x) = \int_{y=-\infty}^{\infty} f(x,y)dy$ for $-\infty < x < \infty$.

$\Longrightarrow$

$$g(x) = \int_{y=-0}^{\infty} k(x+1)e^{-y}\ dy$$
$$= \frac{2}{3}(x+1)\left(\frac{e^{-y}}{-1}\right)_{y=0}^{\infty} = \frac{2}{3}(x+1) \text{ for } 0 < x < 1.$$

The marginal probability distribution of X is, $h(y) = \int_{x=-\infty}^{\infty} f(x,y)dx$ for $-\infty < y < \infty$.

$\Longrightarrow$

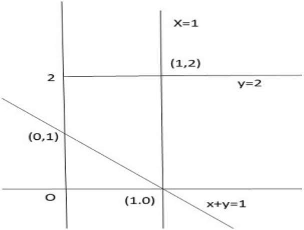
$$h(y) = \int_{x=-0}^{1} k(x+1)e^{-y}\ dx$$
$$= \frac{2}{3}e^{-y}\left(\frac{x^2}{2}+x\right)_{0}^{1} = e^{-y} \text{ for } 0 < y < \infty.$$

Question 3.

Let (X, Y) be the two dimensional random variable with joint probability distribution $f(x,y) = \begin{cases} x^2 + \dfrac{xy}{3}, & \text{if } 0 < x < 1, 0 < y < 2 \\ 0, & \text{else where} \end{cases}$. Then find

1. $P(X + Y \geq 1)$

Solution:



(1)

$$
\begin{aligned}
P(X+Y \geq 1) &= 1 - P(X+y<1) \\
&= 1 - \int_0^1 \int_0^{1-x} f(x,y)\ dydx \\
&= 1 - \int_0^1 \int_0^{1-x} x^2 + \frac{xy}{3}\ dydx \\
&= 1 - \int_0^1 \left(x^2 y + \frac{xy^2}{6} \right)_0^{1-x} dx \\
&= 1 - \int_0^1 \left(x^2(1-x) + \frac{x(1-x)^2}{6} \right) dx
\end{aligned}
$$

$$1 - \frac{7}{72} = \frac{65}{72}.$$

Question 4.

Let (X, Y) be the two dimensional random variable with joint probability distribution $f(x,y) = \begin{cases} e^{-(x+y)}, & \text{if } x \geq 0, y \geq 0 \\ 0, & \text{else where} \end{cases}$. Then find

1. $E(X), E(Y)$ and $E(XY)$.

Solution: We know,

$$\begin{aligned} E(X) &= \int_{x=-\infty}^{\infty} x \left(\int_{y=-\infty}^{\infty} f(x,y)dy \right) dx \\ &= \int_{x=0}^{\infty} x \left(\int_{y=0}^{\infty} f(x,y)dy \right) dx \\ &= \int_{x=0}^{\infty} x \left(\int_{y=0}^{\infty} e^{-(x+y)}dy \right) dx \\ &= \int_{x=0}^{\infty} xe^{-x}dx = 1 \end{aligned}$$

We know,

$$E(Y) = \int_{y=-\infty}^{\infty} y \ \left(\int_{x=-\infty}^{\infty} f(x,y)dx \right) dy$$

$$= \int_{y=0}^{\infty} y \ \left(\int_{x=0}^{\infty} f(x,y)dx \right) dy$$

$$= \int_{y=0}^{\infty} y \ \left(\int_{x=0}^{\infty} e^{-(x+y)}dx \right) dx$$

$$= \int_{y=0}^{\infty} ye^{-y}dx = 1$$

We know,

$$E(XY) = \int_{x=-\infty}^{\infty} \left(\int_{y=-\infty}^{\infty} xyf(x,y)dy \right) dx$$

$$= \int_{x=0}^{\infty} \left(\int_{y=0}^{\infty} xyf(x,y)dy \right) dx$$

$$= \int_{x=0}^{\infty} \left(\int_{y=0}^{\infty} xye^{-(x+y)}dy \right) dx$$

$$
\begin{aligned}
&= \int_{x=0}^{\infty} xe^{-x} \left(\int_{y=0}^{\infty} ye^{-y} dy \right) dx \\
&= \int_{x=0}^{\infty} xe^{-x} dx \times \int_{y=0}^{\infty} ye^{-y} dy = 1.
\end{aligned}
$$

Correlation coefficient

There are, many phenomenae where the changes in one variable are related to the changes in the other variable. When the changes in one variable are associated or followed by changes in the other, is called correlation. If an increase (or decrease) in the values of one variable corresponds to an increase (or decrease) in the other, the correlation is said to be positive. If the increase (or decrease) in one corresponds to the decrease (or increase) in the other, the correlation is said to be negative. If there is no relationship indicated between the variables, they are said to be independent or uncorrelated.

The numerical measure of correlation is called the coefficient of correlation. Let (X, Y) be a two-dimensional random variable. Then the correlation co-efficient between X and Y is denoted by ρ_{XY} and is defined as

$$\rho_{XY} = \frac{E\{(X - E(X))(Y - E(Y))\}}{\sqrt{V(X)V(Y)}}.$$

Note 1.1

$E\{(X - E(X))(Y - E(Y))\}$ is called the covariance of X and Y and is denoted by $\mathrm{COV}(X, Y)$.

Theorem 1.2

Prove that $\rho_{XY} = \dfrac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}}$.

Proof: It is enough to show that
$E\{(X - E(X))(Y - E(Y))\} = E(XY) - E(X)E(Y)$.
Consider
$E\{(X - E(X))(Y - E(Y))\} = E\{XY - XE(Y) - YE(X) + E(X)E(Y)\}$
$= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y)$
$= E(XY) - E(X)E(Y)$.

Note 1.3

If X and Y are independent, then $E(XY) = E(X)E(Y)$. Hence $\rho_{XY} = 0$. The converse need not be true. That is, if $\rho_{XY} = 0$, then X and Y need not be independent.

Consider a random variable X with pdf $f(x) = 1/2, \ -1 \leq x \leq 1$.

Define a random variable $Y = X^2$.

Then $E(XY) - E(X)E(Y) = E(X.X^2) - E(X)E(X^2)$.

$E(X^3) = \int_{-\infty}^{\infty} x^3 f(x)dx = \int_{-1}^{1} x^3 \cdot \frac{1}{2}dx = 0$

$E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_{-1}^{1} x \cdot \frac{1}{2}dx = 0.$

Hence $\rho_{XY} = 0$. But X and Y are not independent.

Theorem 1.4

Prove that $-1 \leq \rho \leq 1$.

Proof: We have $E\{\frac{X-E(X)}{\sqrt{V(X)}} \pm \frac{Y-E(Y)}{\sqrt{V(Y)}}\}^2 \geq 0$

$\Rightarrow E[\frac{(X-E(X))^2}{V(X)} + \frac{(Y-E(Y))^2}{V(Y)} \pm \frac{2(X-E(X))(Y-E(Y))}{\sqrt{V(X)V(Y)}}] \geq 0$

$\Rightarrow \frac{E(X-E(X))^2}{V(X)} + \frac{E(Y-E(Y))^2}{V(Y)} \pm \frac{2E[(X-E(X))(Y-E(Y))]}{\sqrt{V(X)V(Y)}} \geq 0$

$\Rightarrow \frac{V(X)}{V(X)} + \frac{V(Y)}{V(Y)} \pm \frac{2E[(X-E(X))(Y-E(Y))]}{\sqrt{V(X)V(Y)}} \geq 0$

$\Rightarrow 2 \pm 2\rho_{XY} \geq 0$

$\Rightarrow 1 \pm \rho_{XY} \geq 0$

$\Rightarrow 1 + \rho_{XY} \geq 0;\ 1 - \rho_{XY} \geq 0$

$\Rightarrow \rho_{XY} \geq -1;\ \ 1 \geq \rho_{XY};$

$\Rightarrow -1 \leq \rho_{XY} \leq 1.$

Theorem 1.5

If X and Y are linearly related, then show that $\rho_{XY} = \pm 1$.

Proof: Let $Y = a + bX$ with a and b are constants.

We have $\rho_{XY} = \dfrac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}}$

Consider $E(XY) = E(X(a + bX)) = E(aX + bX^2) = aE(X) + bE(X^2)$

$E(Y) = E(a + bX) = a + bE(X)$

$V(Y) = V(a + bX) = b^2 V(X)$

Then $\rho_{XY} = \dfrac{aE(X) + bE(X^2) - E(X)(a + bE(X))}{\sqrt{V(X)b^2 V(X))}}$

$$\rho_{XY} = \frac{aE(X) + bE(X^2) - aE(X) - b(E(X))^2}{\pm bV(X)}$$

$$= \frac{b[E(X^2) - (E(X))^2]}{\pm bV(X)} = \pm 1.$$

Problem 1.6

The random variable (X, Y) has a joint pdf given by $f(x, y) = x + y, 0 \leq x \leq 1,\ 0 \leq y \leq 1$. Compute the correlation co-efficient between X and Y.

Solution: $E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y)dxdy$

$= \int_0^1 \int_0^1 xy(x + y)dxdy = 1/3$

$E(X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y)dxdy = \int_0^1 \int_0^1 x(x + y)dxdy = 7/12$

$E(Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y)dxdy = \int_0^1 \int_0^1 y(x + y)dxdy = 7/12$

$E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2f(x, y)dxdy = \int_0^1 \int_0^1 x^2(x + y)dxdy = 5/12$

$E(Y^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2f(x, y)dxdy = \int_0^1 \int_0^1 y^2(x + y)dxdy = 5/12$

$V(X) = E(X^2) - (E(X))^2 = 11/144$ and

$V(Y) = E(Y^2) - (E(Y))^2 = 11/144$.

$$\rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{V(X)V(Y)}} = \frac{-1}{11}.$$

Problem 1.7

Show that $V(aX+bY)=a^2V(X)+b^2V(Y)+2abCOV(X,Y)$

Solution: $V(aX+bY)=E[(aX+bY)-E(aX+bY)]^2$
$=E[aX+bY-aE(X)-bE(Y)]^2$
$=E[a(X-E(X))+b(Y-E(Y)]^2$
$=E[a^2(X-E(X))^2+b^2(Y-E(Y))^2+2ab(X-E(X))(Y-E(Y))]$
$=a^2E(X-E(X))^2+b^2E(Y-E(Y))^2+2abE\{(X-E(X))(Y-E(Y))\}$
$=a^2V(X)+b^2V(Y)+2abCOV(X,Y).$

Note 1.8

If X and Y are independent random variables, then $E(XY) = E(X)E(Y)$. Hence $COV(X, Y) = 0$. Therefore $V(aX + bY) = a^2 V(X) + b^2 V(Y)$.

Problem 1.9

If X_1, X_2 and X_3 are uncorrelated random variables having the same standard deviation. Then find the correlation co-efficient between $X_1 + X_2$ and $X_2 + X_3$.

Solution: Since X_1, X_2 and X_3 are uncorrelated, then $\rho_{X_1X_2} = 0$, $\rho_{X_2X_3} = 0$ and $\rho_{X_3X_1} = 0$.
Let $V(X_1) = V(X_2) = V(X_3) = \sigma$.
Now, take $U = X_1 + X_2$ and $V = X_2 + X_3$. Then

$$\rho_{UV} = \frac{E(UV) - E(U)E(V)}{\sqrt{V(U)V(V)}}$$

$$= \frac{E((X_1 + X_2)(X_2 + X_3)) - E(X_1 + X_2)E(X_2 + X_3)}{\sqrt{V(X_1 + X_2)V(X_2 + X_3)}}$$

$$= \frac{E(X_1X_2 + X_1X_3 + X_2^2 + X_2X_3) - (E(X_1) + E(X_2))(E(X_2) + E(X_3))}{\sqrt{V(X_1 + X_2)V(X_2 + X_3)}}$$

$$= \frac{E(X_1X_2) + E(X_1X_3) + E(X_2^2) + E(X_2X_3) - E(X_1)E(X_2) - E(X_1)E(X_3) - (E(X_2))^2 - (E(X_2)E(X_3))}{\sqrt{(V(X_1) + V(X_2))(V(X_2) + V(X_3))}}$$

$[V(X_1 + X_2) = V(X_1) + V(X_2) + 2COV(X_1, X_2)$. Since $\rho_{X_1X_2} = 0$, then $COV(X_1, X_2) = 0]$

$$= \frac{E(X_2^2) - (E(X_2))^2}{\sqrt{2\sigma \cdot 2\sigma}}$$

$$= \frac{V(X_2)}{\sqrt{2\sigma \cdot 2\sigma}} = \frac{\sigma}{2\sigma} = 1/2.$$

Lines of Regression

It frequently happens that the dots of the scatter diagram generally, tend to cluster along a well defined direction which suggests a linear relationship between the variables x and y. Such a line of best-fit for the given distribution of dots is called the line of regression. There are two such lines, one line gives the best possible mean values of y for each specified value of x is known as the line of regression of y on x and the other line gives the best possible mean values of x for given values of y is known as the line of regression of x on y.

Consider first the line of regression of y on x. Let the straight line satisfying the general trend of n dots in a scatter diagram be $y = a + bx$......(1)
We have to determine the constants a and b so that (1) gives for each value of x, the best estimate for the average value of y in accordance with the principle of least squares, then the normal equations for a and b are
$\Sigma y = na + b\Sigma x$ ....(2) and
$\Sigma xy = a\Sigma x + b\Sigma x^2$.......(3)
(2) gives $\frac{1}{n}\Sigma y = a + b.\frac{1}{n}\Sigma x$
$\Rightarrow \bar{y} = a + b\bar{x}$.
This shows that means of x and y $(\bar{x}, \bar{y})$ lie on (1).

By shifting the origin to $(\bar{x}, \bar{y})$, (3) takes the form
$\Sigma(x-\bar{x})(y-\bar{y}) = a\Sigma(x-\bar{x}) + b\Sigma(x-\bar{x})^2$, but $a\Sigma(x-\bar{x}) = 0$,

$$\therefore b = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{\Sigma XY}{\Sigma X^2} = \frac{\Sigma XY}{n\sigma_x^2} = r\frac{\sigma_y}{\sigma_x}$$

Thus the line of best fit becomes $y - \bar{y} = r\dfrac{\sigma_y}{\sigma_x}(x - \bar{x})$
which is the equation of the line of regression of y on x. Its slope is called the regression coefficient of y on x.
Interchanging x and y, we find that the line of regression of x on y is

$$x - \bar{x} = r\frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

Thus the regression coefficient of y on x $= r\sigma_y/\sigma_x$ and the regression coefficient of x on y $= r\sigma_x/\sigma_y$.

Problem 2.1

The two regression equations of the variables x and y are $x = 19.13 - 0.87y$ and $y = 11.64 - 0.50x$. Find (i) mean of x's, (ii) mean of y's and (iii) the correlation coefficient between x and y.

Solution: Since the mean of x's and the mean of y's lie on the two regression lines, we have

$\bar{x} = 19.13 - 0.87\bar{y}$ ......(i)

$\bar{y} = 11.64 - 0.50\bar{x}$..........(ii)

Multiplying (ii) by 0.87 and subtracting from (i), we have

$[1 - (0.87)(0.50)]\bar{x} = 19.13 - (11.64)(0.87)$ or $0.57\bar{x} = 9.00$ or $\bar{x} = 15.79$

$\bar{y} = 11.64 - (0.50)(15.79) = 3.74$

$\therefore$ regression coefficient of y on x is -0.50 and that of x on y is -0.87.

Since the coefficient of correlation is the geometric mean between the two regression coefficients, then

$r = \sqrt{(-0.50)(-0.87)} = \sqrt{0.43} = -0.66$

(negative sign is taken since both the regression coefficients are negative).

Problem 2.2

Calculate the lines of regression of y on x and x on y by using the following data.

x	28	25	24	29	31	22	21	25	26	28
y	22	23	21	25	26	20	19	21	21	24

Solution: $\bar{x} = \dfrac{\Sigma x_i}{n} = 25.9$; $\bar{y} = \dfrac{\Sigma y_i}{n} = 22.2$

x	y	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
28	22	2.1	-0.2	4.41	0.04	-0.42
25	23	-0.9	0.8	0.81	0.64	-0.72
24	21	-1.9	-1.2	3.61	1.44	2.28
29	25	3.1	2.8	9.61	7.84	8.68
31	26	5.1	3.8	26.01	14.44	19.38
22	20	-3.9	-2.2	15.21	4.84	8.58
21	19	-4.9	-3.2	24.01	10.24	15.68
25	21	-0.9	-1.2	0.81	1.44	1.08
26	21	0.1	-1.2	0.01	1.44	-0.12
28	24	2.1	1.8	4.41	3.24	3.78

$\Sigma X^2 = 88.9; \quad \Sigma Y^2 = 45.6; \;\; \Sigma XY = 58.2$

The regression line of x on y is $x - \bar{x} = r\dfrac{\sigma_x}{\sigma_y}(y - \bar{y})$

Now, $r\dfrac{\sigma_x}{\sigma_y} = \dfrac{\Sigma XY}{n\sigma_x\sigma_y} \times \dfrac{\sigma_x}{\sigma_y} = \dfrac{\Sigma XY}{n{\sigma_y}^2} = \dfrac{\Sigma XY}{n\dfrac{\Sigma Y^2}{n}} = \dfrac{\Sigma XY}{\Sigma Y^2}$

Hence the regression line of x on y is $x - \bar{x} = \dfrac{\Sigma XY}{\Sigma Y^2}(y - \bar{y})$

$x - 25.9 = \dfrac{58.2}{45.6}(y - 22.2)$

$\Rightarrow x - 25.9 = 1.2763(y - 22.2)$

$\Rightarrow x = 1.2763y - 2.44.$

The regression line of y on x is $y - \bar{y} = \dfrac{\Sigma XY}{\Sigma X^2}(x - \bar{x})$

$y - 22.2 = \dfrac{58.2}{88.9}(x - 25.9)$

$\Rightarrow y = 5.244 + 0.6547x.$